

Topic Name:- File System Automation

Team Name:- TrailBlazers

Roles

☐ **Scrum master**

Bhagyashree Vijay Wagh

☐ **Developers**

Komal Anil Mahajan

Mamata Kiran Chaudhari

Srushti Subhash Mahajan

☐ **Testers**

Ayush Ajit Baviskar

Vishal Shivaji Gadekar

Project Objective

- Automates file and folder management to reduce manual effort and improve efficiency.
- Organizes files into appropriate folders based on predefined rules such as file type, extension, or file name.
- Monitors selected directories continuously and performs actions automatically when changes are detected.
- Reduces human errors by maintaining a well-structured and organized file system.
- Saves time by simplifying repetitive file management tasks and increasing productivity.
- Provides a reliable, secure, and user-friendly solution for efficient file system management.

Features

1. Face Recognition
2. Automatic Backup System
3. Smart File Search
4. Folder Synchronization
5. Duplicate File Management
6. Automatic File Classification
7. File Version Control
8. File Recovery System
9. Storage Optimization
10. Automatic Report Generation
11. File Password Protection

Future Scope

1. File Access Analytics Dashboard

1. Face Recognition

Purpose:

To provide secure access by allowing only authorized users to use the system.

Functional Requirement:

- Accept a reference image from the user.
- Store the reference image in the system.
- Scan and compare images with the reference image.
- Identify all images that match the reference image.
- Save all matching images in a separate folder.
- Display the number of matching images found.
- Allow the user to update the reference image.
- Keep a record of the image matching process.

2. Automatic File Classification

Purpose:

To automatically organize files into appropriate folders based on their type.

Functional Requirement:

- Classify files based on their type.
- Move files to appropriate folders automatically.
- Support multiple file formats.
- Organize folders efficiently.
- Create folders automatically.
- Rename files if needed.
- Show classified file list.

3.Smart File Search

Purpose:

To quickly locate files using different search criteria.

Functional Requirement:

- Search files by name.
- Search files by extension.
- Search files by date.
- Search files by size.
- Display search results instantly.

4. Folder Synchronization

Purpose:

To keep files updated and consistent across multiple folders.

Functional Requirement:

- Synchronize files between two folders.
- Automatically update modified files.
- Copy newly added files.
- Maintain consistency between folders.
- Sync files automatically.
- Show sync status.
- Update changed files.

5. Automatic Backup System

Purpose:

To protect important files by creating backups before any modification.

Functional Requirement:

- Create backup before file modification.
- Store backup copies securely.
- Restore files from backup.
- Schedule automatic backups.
- Check if backup is successful.
- Delete old backups.
- Show backup completed message.

6. File Recovery System

Purpose:

To recover accidentally deleted files and prevent data loss.

Functional Requirement:

- Recover deleted files.
- Restore files from backup.
- Display recoverable files.
- Permanently delete unwanted files.
- Preview deleted files.
- Recover multiple files.
- Show recovery status.

7. Duplicate File Management

Purpose:

To identify and remove duplicate files, reducing storage usage and improving file organization.

Functional Requirement:

- Detect duplicate files.
- Compare duplicate file names and sizes.
- Remove duplicate files.
- Generate duplicate file reports.
- Show duplicate files before deleting.
- Move duplicate files to another folder.
- Show storage space saved.

8. File Version Control

Purpose:

To maintain previous versions of files and restore them whenever required.

Functional Requirement:

- Create a new version after every file modification.
- Maintain previous versions of files.
- Restore older versions when required.
- Track version history.
- Save every new version of the file.
- Compare old and new versions.
- Delete old versions if not needed.

9. Storage Optimization

Purpose:

To optimize storage by removing unnecessary files and improving disk usage.

Functional Requirement:

- Detect duplicate files.
- Identify large files.
- Find unused files.
- Display storage usage statistics.
- Free up unnecessary storage space.
- Find temporary files.
- Show free storage space.
- Suggest files to delete.

10. Automatic Report Generation

Purpose:

To maintain a detailed record of all file operations for monitoring and analysis.

Functional Requirement:

- Record all file operations.
- Generate daily and weekly reports.
- Export reports in PDF or CSV format.
- Display operation history.
- Create monthly reports.
- Save reports automatically.
- Show report summary.

11. File Password Protection

Purpose:

To protect sensitive files by allowing access only to users who enter the correct password, ensuring data security and preventing unauthorized access.

Functional Requirement:

- Set a password for selected files.
- Verify the password before opening the file.
- Encrypt the file for enhanced security.
- Allow only authorized users to access protected files.
- Change or remove the file password.
- Display an error message for incorrect passwords.
- Maintain a log of password-protected file access.

Future Scope:-

1. File Access Analytics Dashboard

Purpose:

To monitor and analyze file usage by tracking file access history, helping users understand file activity and improve file management.

Functional Requirement:

- Track file access history.
- Count how many times each file is opened.
- Display the last accessed date and time.
- Identify the most frequently accessed files.
- Display file usage statistics.
- Generate file access reports.
- Export analytics reports in PDF or CSV format.

❑ Task Distribution to Developers

Developers	Task 1	Task 2	Task 3	Task 4	Working Hours
1. Komal Anil Mahajan	• Face Recognition (4 hrs)	• Automatic Backup System (3 hrs)	• Duplicate File Management (2 hrs)		Total :- 9 hrs
2. Mamata Kiran Chaudhari	• File Version Control (3 hrs)	• Folder Synchronization (2 hrs)	• File Password Protection (2 hrs)	• File Recovery System (2 hrs)	Total :- 9 hrs
3. Srushti Subhash Mahajan	• Storage Optimization (3 hrs)	• Automatic Report Generation (2 hrs)	• Automatic File Classification (2 hrs)	• Smart File Search (2 hrs)	Total :- 9 hrs

☐ Task Distribution to Testers

Testers	Task 1	Task 2	Task 3	Task 4	Task 5	Task 6	Working Time
1. Ayush Ajit Baviskar	• Face Recogniti on (2 hrs)	• Automatic Backup System (30 min)	• File Password Protectio n (30 min)	• File Recovery System (30 min)	• Storage Optimizati on (30 min)		Total :- 4 hrs
2. Vishal Shivaji Gadekar	• File Version Control (1 hrs)	• Folder Synchroni zation (30 min)	• Smart File Search (30 min)	• Duplicate File Managem ent (30 min)	• Automatic File Classificat ion (1 hrs)	• Automatic Report Generatio n (30 min)	Total :- 4 hrs

DataBase Table

Requirement	Table Name	Column Names
1. Face Recognition	Face_Recognition	reference_id,image_name,image_path,upload_date,scan_date,matched_images_count,matching_folder_path,match_status
2. Automatic Backup System	Backup_History	backup_id, file_id, file_name, backup_location, backup_type, backup_date, backup_status
3. File Version Control	File_Version	version_id, file_id, file_name, version_number, modified_by, modified_date, version_notes
4. File Recovery System	Recycle_Bin	recovery_id, file_id, file_name, deleted_date, recovered_date, recovery_status, deleted_by
5. Automatic Report Generation	Generated_Reports	report_id, report_name, report_type, generated_by, generated_date, report_status

Acceptance Criteria

- System authenticates registered users and blocks unauthorized users.
- A new version is created after every file modification and older versions can be restored.
- Files are synchronized correctly between selected folders.
- Duplicate and large files are detected and storage statistics are displayed.
- Reports are generated correctly and exported to PDF/CSV.
- Files are automatically moved to the correct folders based on file type.
- Backup is created before file modification and can be restored successfully.
- Files are searched correctly by name, extension, date, and size.
- Deleted files can be recovered or permanently deleted.
- Duplicate files are detected and removed successfully.
- Protected files require the correct password before opening.

Thank You!